

### Advanced (Habanero)

- Q1.** What is the purpose of a scatter plot?
- (A) To display categorical data
  - (B) To show the relationship between two variables
  - (C) To illustrate a linear equation
  - (D) To graph a quadratic function
  - (E) To visualize a system of equations
- Q2.** Which type of correlation is shown by the data points (1, 2), (2, 4), (3, 6)?
- (A) Positive
  - (B) Negative
  - (C) None
  - (D) Weak
  - (E) Strong
- Q3.** What is the equation of the line of best fit for the data points (0, 0), (1, 2), (2, 3)?
- (A)  $y = x$
  - (B)  $y = 2x$
  - (C)  $y = x + 1$
  - (D)  $y = 2x - 1$
  - (E)  $y = x^2$
- Q4.** If a scatter plot shows a strong positive correlation, what can be predicted?
- (A) As  $x$  increases,  $y$  decreases
  - (B) As  $x$  increases,  $y$  increases
  - (C) As  $x$  decreases,  $y$  increases
  - (D) As  $x$  decreases,  $y$  decreases
  - (E) No prediction can be made
- Q5.** Which of the following is a characteristic of the line of best fit?
- (A) It passes through all data points
  - (B) It is a vertical line
  - (C) It is a horizontal line
  - (D) It minimizes the sum of the squared errors
  - (E) It maximizes the sum of the squared errors
- Q6.** What is the purpose of the line of best fit?
- (A) To predict future values
  - (B) To summarize the data
  - (C) To display the data
  - (D) To find the correlation
  - (E) To graph a quadratic function
- Q7.** If a scatter plot shows no correlation, what can be concluded?
- (A) There is a cause-and-effect relationship
  - (B) There is no cause-and-effect relationship
  - (C) The data is invalid
  - (D) The data is not relevant
  - (E) No conclusion can be made
- Q8.** Which type of data is best represented by a scatter plot?
- (A) Categorical data
  - (B) Quantitative data
  - (C) Qualitative data
  - (D) Discrete data
  - (E) Continuous data

#### Open Ended Questions (FRQ)

- Q9.** Create a scatter plot for the data points (1, 2), (2, 3), (3, 5) and draw the line of best fit. Use the line of best fit to predict the value of  $y$  when  $x = 4$ .
- Q10.** A company finds that the number of hours worked ( $x$ ) and the amount of money earned ( $y$ ) are related. The data points are (2, 10), (4, 20), (6, 30). Create a scatter plot and draw the line of best fit. Use the line of best fit to predict the amount of money earned when 8 hours are worked.

#### Chef's Tips

- Tip 1: Emphasize the importance of creating accurate scatter plots
- Tip 2: Encourage students to use technology to find the line of best fit
- Tip 3: Remind students to check their work by plugging in values